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Cold plate and manifold integration for high reliability

Abstract

A device package comprising an integrated cooling assembly. The device package comprises a semiconductor device attached to a substrate and a cold plate attached to the semiconductor device. The cold plate's footprint is larger than the footprint of the semiconductor device so one or more portions of the cold plate extends past the edges of the semiconductor device's footprint. One or more standoffs are disposed adjacent to the semiconductor device to support the one or more portions of the cold plate that extend past the edges of the semiconductor device's footprint.

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Background/Summary

BACKGROUND

(1) The present disclosure relates to advanced packaging for microelectronic devices, and in particular, embedded cooling systems for device packages and methods of manufacturing the same.

SUMMARY

(2) Energy consumption poses a critical challenge for the future of large-scale computing as the world's computing energy requirements are rising at a rate that most would consider unsustainable. Some models predict that the information, communication and technology (ICT) ecosystem could exceed 20% of global electricity use by 2030, with direct electrical consumption by large-scale computing centers accounting for more than one-third of that energy usage. A significant portion of the energy used by such large-scale computing centers is devoted to cooling, since even small increases in operating temperatures can negatively impact the performance of microprocessors, memory devices, and other electronic components. While some of this energy is expended to operate the cooling systems that are directly cooling the chips (e.g., heat spreaders, heat pipes, etc.), energy consumption/costs for indirect cooling can also be quite staggering. Indirect cooling energy costs include, for example, cooling or air conditioning of data center buildings. Data center buildings can house thousands, to tens of thousands or more, of high-performance chips in server racks, and each of those high-performance chips is a heat source. An uncontrolled ambient temperature in a data center will adversely affect the performance of the individual chips, and the data center system performance as a whole.

(3) Thermal dissipation in high-power density chips (semiconductor devices/dies) is also a critical challenge as improvements in chip performance (e.g., through increased gate or transistor density due to advanced processing nodes, evolution of multi-core microprocessors, etc.) have resulted in increased power density and a corresponding increase in thermal flux that contributes to elevated chip temperatures. Higher density of transistors also increases the length of metal wiring on the chips, which generates its own additional thermal flux due to Joule heating of these wires due to higher currents. These elevated temperatures are undesirable as they can degrade the chip's operating performance, efficiency, reliability, and amount of remaining life. Cooling systems used to maintain the chip at a desired operating temperature typically remove heat using one or more heat dissipation devices (e.g., thermal spreaders, heat pipes, cold plates, liquid cooled heat pipe systems, thermal-electric coolers, heat sinks, etc.). One or more thermal interface material(s) (e.g., thermal paste, thermal adhesive, and/or thermal gap filler), may be used to facilitate heat transfer between the surfaces of a chip and heat dissipation device(s). A thermal interface material(s) (TIM(s)) is any material that is inserted between two components to enhance the thermal coupling therebetween. Unfortunately, the combined thermal resistance of (i) the thermal resistance of interfacial boundary regions between a TIM(s) and the chip and/or the heat dissipation device(s), and (ii) the thermal resistance of a thermal interface material(s) itself can inhibit heat transfer from the chip to the heat dissipation devices, undesirably reducing the cooling efficiency of the cooling system.

(4) Generally speaking, there are multiple components between the heat dissipating sources (i.e., active circuitry) in the chips and the heat dissipation devices, each of which contributes to the system thermal resistance cumulatively along the heat transfer paths and raises chip junction temperatures from the ambient.

(5) Such cooling systems can suffer from reduced cooling efficiency due to the design and manufacture of system components.

(6) Accordingly, there exists a need in the art for improved energy-efficient cooling systems, by reducing system thermal resistance, and methods of manufacturing the same.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The above and other objects and advantages of the disclosure will be apparent upon consideration of the following detailed description, taken in conjunction with the accompanying drawings, in which:

(2) FIG. 1 illustrates a device package with an external heat sink, in accordance with some embodiments of the disclosure;

(3) FIG. 2A is a schematic plan view of an example of a system panel, in accordance with some embodiments of the present disclosure;

(4) FIG. 2B is a schematic partial sectional side view of a device package mounted on a PCB, in accordance with some embodiments of the present disclosure;

(5) FIG. 2C is a schematic exploded isometric view of the device package in FIG. 2B, in accordance with some embodiments of the present disclosure;

(6) FIG. 3 is a schematic sectional view of an example device package that may be used with the system panel, in accordance with some embodiments of the present disclosure;

(7) FIG. 4 is a schematic sectional view of an integrated cooling assembly of the device package, in accordance with some embodiments of the present disclosure;

(8) FIG. 5 is a schematic sectional view of another example device package that may be used with the system panel, in accordance with some embodiments of the present disclosure;

(9) FIG. 6 shows a method that can be used to manufacture the device package, in accordance with some embodiments of the present disclosure;

(10) FIG. 7 shows a schematic top view of a portion of a device package that may be used with the system panel, in accordance with some embodiments of the present disclosure;

(11) FIG. 8 shows a schematic sectional view of a portion of a device package that may be used with the system panel, in accordance with some embodiments of the present disclosure;

(12) FIG. 9 shows a schematic sectional view of a portion of a device package that may be used with the system panel, in accordance with some embodiments of the present disclosure;

(13) FIG. 10 shows a schematic top view of a portion of a device package that may be used with the system panel, in accordance with some embodiments of the present disclosure;

(14) FIG. 11 shows a schematic top view of a portion of a device package that may be used with the system panel, in accordance with some embodiments of the present disclosure;

(15) FIG. 12 shows a schematic sectional view of a portion of a device package that may be used with the system panel, in accordance with some embodiments of the present disclosure;

(16) FIG. 13 shows a schematic sectional view of a portion of a device package that may be used with the system panel, in accordance with some embodiments of the present disclosure;

(17) FIG. 14 shows a schematic sectional view of a portion of a device package that may be used with the system panel, in accordance with some embodiments of the present disclosure;

(18) FIG. 15 shows a schematic sectional view of a portion of a device package that may be used with the system panel, in accordance with some embodiments of the present disclosure;

(19) FIG. 16 shows a schematic sectional view of a portion of a device package that may be used with the system panel, in accordance with some embodiments of the present disclosure;

(20) FIG. 17 shows a schematic sectional view of a portion of a device package that may be used with the system panel, in accordance with some embodiments of the present disclosure;

(21) FIGS. 18A-18E show a method of manufacturing a portion of a device package that may be used with the system panel, in accordance with some embodiments of the present disclosure; and

(22) FIGS. 19A-19D show a method of manufacturing a portion of a device package that may be used with the system panel, in accordance with some embodiments of the present disclosure.

(23) The figures herein depict various embodiments of the present disclosure for purposes of illustration only. It will be appreciated that additional or alternative structures, assemblies, systems, and methods may be implemented within the principles set out by the present disclosure.

DETAILED DESCRIPTION

(24) As used herein, the term “substrate” means and includes any workpiece, wafer, or article that provides a base material or supporting surface from which or upon which components, elements, devices, assemblies, modules, systems, or features of the heat-generating devices, packaging components, and cooling assembly components described herein may be formed or mounted. The term “substrate” also includes “semiconductor substrates” that provide a supporting material upon which elements of a semiconductor

device are fabricated or attached, and any material layers, features, and/or electronic devices formed thereon, therein, or therethrough. Examples of substrate material that may be used in applications that generate high thermal density include, but are not limited to, Si, GaN, SiC, InP, GaP, InGaN, AlGaInP, AlGaAs, etc.

(25) As described below, the semiconductor substrates herein generally have a “device side,” (e.g., the side on which semiconductor device elements are fabricated, such as transistors, resistors, and capacitors) and a “backside” that is opposite the device side. The term “active side” should be understood to include a surface of the device side of the substrate and may include the device side surface of the semiconductor substrate and/or a surface of any material layer, device element, or feature formed thereon or extending outwardly therefrom, and/or any openings formed therein. Thus, it should be understood that the material(s) that forms the active side may change depending on the stage of device fabrication and assembly. Similarly, the term “non-active side” (opposite the active side) includes the non-active side of the substrate at any stage of device fabrication, including the surfaces of any material layer, any feature formed thereon, or extending outwardly therefrom, and/or any openings formed therein. Thus, the terms “active side” or “non-active side” may include the respective surfaces of the semiconductor substrate at the beginning of device fabrication and any surfaces formed during material removal, (e.g., after substrate thinning operations). Depending on the stage of device fabrication or assembly, the terms “active sides” and “non-active sides” are also used to describe surfaces of material layers or features formed on, in, or through the semiconductor substrate, whether or not the material layers or features are ultimately present in the fabricated or assembled device. For example, in some instances, the term “active side” is used to indicate a surface of a substrate that will in the future, but does not yet, include semiconductor device elements.

(26) Spatially relative terms are used herein to describe the relationships between elements, such as the relationships between substrates, heat-generating devices, cooling assembly components, device packaging components, and other features described below. Unless the relationship is otherwise defined, terms such as “above,” “over,” “upper,” “upwardly,” “outwardly,” “on,” “below,” “under,” “beneath,” “lower,” “top,” “bottom” and the like are generally made with reference to the X, Y, and Z directions set forth by X, Y and Z axes in the drawings. Thus, it should be understood that the spatially relative terms used herein are intended to encompass different orientations of the substrate and, unless otherwise noted, are not limited by the direction of gravity. Unless the relationship is otherwise defined, terms describing the relationships between elements such as “disposed on,” “embedded in,” “coupled to,” “connected by,” “attached to,” “bonded to,” and the like, either alone or in combination with a spatially relevant term, include both relationships with intervening elements and direct relationships where there are no intervening elements. Furthermore, the term “horizontal” is generally made with reference to the X-axis direction and the Y-axis direction set forth in the drawings. The term “vertical” is generally made with reference to the Z-axis direction set forth in the drawings.

(27) Various embodiments disclosed herein include bonded structures in which two or more elements are directly bonded to one another without an intervening adhesive (referred to herein as “direct bonding,” “direct dielectric bonding,” or “directly bonded”). In some embodiments, direct bonding includes the bonding of a single material on the first of the two or more elements and a single material on a second one of the two or more elements, where the single material on the different elements may or may not be the same. For example, bonding a layer of one inorganic dielectric (e.g., silicon oxide) to another layer of the same or different inorganic dielectric. As discussed in more detail below, the process of direct bonding (e.g., direct dielectric bonding) provides a reduction of thermal resistance between a semiconductor device and a cold plate. Examples of dielectric materials used in direct bonding include oxides, nitrides, oxynitrides, carbonitrides, and oxycarbonitrides, etc., such as, for example, silicon oxide, silicon nitride, silicon oxynitride, silicon carbonitride, silicon oxycarbonitride, etc. Direct bonding can also include bonding of multiple materials on one element to multiple materials on the other element (e.g., hybrid bonding). As used herein, the term “hybrid bonding” refers to a species of direct bonding having both i) at least one (first) nonconductive feature directly bonded to another (second) nonconductive feature, and ii) at least one (first) conductive feature directly bonded to another (second) conductive feature, without any intervening adhesive. The resultant bonds formed by this technique may be described as “hybrid bonds” and/or “direct hybrid bonds.” In some hybrid bonding embodiments, there are many first conductive features, each directly bonded to a second conductive feature, without any intervening adhesive. In some embodiments, nonconductive features on the first element are directly bond to nonconductive features of the second

element at room temperature without any intervening adhesive, which is followed by bonding of conductive features of the first element directly bonded to conductive features of the second element via annealing at slightly higher temperatures (e.g., >100° C., >200° C., >250° C., >300° C., etc.)

(28) Unless otherwise noted, the terms “cooling assembly” and “integrated cooling assembly” generally refer to a semiconductor device and a cold plate attached to the semiconductor device. Typically, the cold plate is formed with recessed surfaces that define one or more fluid cavities (e.g., coolant chamber volume(s) or coolant channel(s)) between the cold plate and the semiconductor device. In embodiments where the cold plate is formed with plural fluid cavities, each fluid cavity may be defined by cavity dividers and/or sidewalls of the cold plate. For example, cavity dividers may be spaced apart from each other and extend laterally between opposing cold plate sidewalls (e.g., in one direction between a first pair of opposing cold plate sidewalls, or in two directions between orthogonal pairs of opposing cold plate sidewalls). The cavity dividers and the cold plate sidewalls may collectively define adjacent fluid cavities therebetween. The cold plate may comprise a polymer material.

(29) The cold plate may be attached to the semiconductor device by use of a compliant adhesive layer or by direct bonding or hybrid bonding. Direct bonding may include direct dielectric bonding techniques as described herein, and may give rise to direct dielectric bonds. Hybrid bonding may include hybrid bonding techniques as described herein, and may give rise to direct hybrid bonds. For example, the cold plate may include material layers and/or metal features that facilitate direct bonding or hybrid bonding with the semiconductor device. Beneficially, the backside of the semiconductor device is directly exposed to coolant fluids flowing through the integrated cooling assembly, thus providing for direct heat transfer therebetween. Unless otherwise noted, the integrated cooling assemblies described herein may be used with any desired fluid (e.g., liquid, gas, and/or vapor-phase coolants, such as water, glycol, etc.). In some embodiments, the coolant fluid(s) may contain additives to enhance the conductivity of the coolant fluid(s) within the integrated cooling assemblies. The additives may comprise, for example, nano-particles of carbon nanotubes, nano-particles of graphene, and/or nano-particles of metal oxides. The concentration of these nano-particles may be less than 1%, less than 0.2%, or less than 0.05%. The coolant fluids may also contain a small amount of glycol or glycols (e.g., propylene glycol, ethylene glycol, etc.) to reduce frictional shear stress and drag coefficient in the coolant fluid(s) within the integrated cooling assembly.

(30) Exemplary fluids available for use in the various thermal solution embodiments include: water (either purified or deionized), a glycol (e.g., ethylene glycol, propylene glycol), glycols mixed with water (e.g., ethylene glycol mixed with water (EGW) or propylene glycol mixed with water (PGW)), dielectric fluids (e.g. fluorocarbons, polyalphaolefin (PAO), isoparaffins, synthetic esters, or very high viscosity index (VHVI) oils), or mineral oils. Additionally, depending upon design and operating conditions, these fluids may be used in single-phase liquid, single-phase vapor, two-phase liquid/vapor or two-phase solid/liquid. All of these fluids and fluid mixtures will alter the thermohydraulic and heat transfer properties by altering the temperatures where phase change occurs, as well as meeting design temperature and pressure conditions for the component being cooled or warmed and the thermal solution being deployed. Additionally, multiple combinations of the fluid phases may be employed in various hybrid configurations to meet the particular cooling or warming needs of a respective implementation and still be within the scope of the contemplated embodiments.

(31) Additionally, in some embodiments part or all the cooling is provided by gases. Exemplary gases include atmospheric air and/or one or more inert gases such as nitrogen. Atmospheric air may be taken to mean the mixture of different gases in Earth's atmosphere made up of about 78% nitrogen and 21% oxygen.

(32) Depending on the design needs of a thermal solution system using the disclosed embodiments, engineered dielectric cooling fluids may be used. Some examples of dielectric fluids used for cooling semiconductors include: 3M™ Fluorinert™ Liquid FC-40—A non-flammable, dielectric fluid that can be used in direct contact with live electronics; 3M™ Novec™ Engineered Fluids—A non-flammable, dielectric fluid that can be used in direct contact with live electronics; Galden® PFPE (perfluoropolyether) products used as heat transfer fluids; EnSolv Fluoro HTF—A solvent with a high boiling point and low pour point that can be used for semiconductor wafer cooling. It is understood that in the selection of the cooling fluid, system design aspects such as operating temperatures and pressures, fluid flow rates, fluid viscosity, and other properties will require evaluation when selecting the appropriate cooling fluid.

(33) In some embodiments, the cooling fluids may contain microparticles and/or nanoparticle additives to enhance the conductivity of the cooling fluid within the integrated cooling assemblies. Choi and Eastman

(1995) from Argonne National Laboratory, U.S.A. (Yu et al., 2007) coined the word “nanofluid.” Nanofluids are engineered fluids prepared by suspending the nano-sized (1-100 nm) particles of metals/non-metals and their oxide(s) with a base/conventional fluid. The suspension of high thermal conductivity metals/non-metals and their oxides nanoparticles enhances the thermal conductivity and heat transfer ability, etc. of the base fluid. The additives to the underlying cooling fluid may comprise for example, nanoparticles of carbon nanotube, nanoparticles of graphene, or nanoparticles of metal oxides. When the cooling fluid contains microparticles, the microparticles are typically 10 microns or less in diameter. Silicon oxide microparticles may be used.

(34) The volume concentration of these micro or nanoparticles may be less than 1%, less than 0.2%, or less than 0.05%. Depending upon the liquid and micro/nanoparticle type chosen for the cooling fluid, higher volume concentrations of 10% or less, 5% or less, or 2% or less may be used. The cooling fluids may also contain small amounts of glycol or glycols (e.g., propylene glycol, ethylene glycol etc.) to reduce frictional shear stress and drag coefficient in the cooling fluid within the integrated cooling assembly. The availability of different base fluids (e.g., water, ethylene glycol, mineral or other stable oils, etc.) and different nanomaterials provide a variety of nanomaterial options for nanofluid solutions to be used in the various embodiments. These nanomaterial option groups such as aforementioned metals (e.g., Cu, Ag, Fe, Au, etc.), metal oxides (e.g., TiO_2 , Al_2O_3 , CuO , etc.), carbons (e.g., CNTS, graphene, diamond, graphite . . . etc.), or a mixture of different types of nanomaterials. Metal nanoparticles (Cu, Ag, Au . . .), metal oxide nanoparticles (Al_2O_3 , TiO_2 , CuO), and carbon-based nanoparticles are commonly employed elements. Silicon oxide nanoparticles may also be used. Using cooling fluids with micro and/or nanoparticles when practicing the various embodiments disclosed herein can result in increased heat removal efficiencies and effectiveness.

(35) The fluid control design aspects of specific embodiments may require the nanofluids to be magnetic to facilitate either movement or cessation of movement of the fluids within the semiconductor structures. Magnetic nanofluids (MNFs) are suspensions of a non-magnetic base fluid and magnetic nanoparticles. Magnetic nanoparticles may be coated with surfactant layers such as oleic acid to reduce particle agglomeration and/or settling. Magnetic nanoparticles used in MNFs are usually made of metal materials (ferromagnetic materials) such as iron, nickel, cobalt, as well as their oxides such as spinel-type ferrites, magnetite (Fe_3O_4), and so forth. The magnetic nanoparticles used in MNFs typically range in size from about 1 to 100 nanometers (nm).

(36) This disclosure describes embodiments involving the architecture of system and component elements that can be employed to provide for the cooling of semi-conductor components, packaging, and boards. However, those skilled in the art will appreciate the disclosed components and arrangements can be deployed and used in scenarios where component heat up or thermal warm up is desired for a component that is currently outside the low end of the desired operational range. Components that are outside the low end of their operational range can, if started in a cold environment, experience thermal warping or cracking up to and including thermal overexpansion and contact separation that may impair the successful operation of the system. Therefore, in these scenarios, the architectures and embodiments disclosed herein can be used where the indirect thermal solutions supporting them are repurposed or operated in a hybrid configuration to provide warming fluids or heat transfer media to accomplish the warm-up or heat-up scenario. These scenarios are controlled by systems not shown here to bring temperatures up at a speed or timing that enables the materials to avoid the excessive thermal expansion or unequal thermal expansion that may occur among the materials of the semiconductor or packaging being serviced by the thermal solution. Once the component or packaging is brought up into the normal operating range, it can be safely started and brought to a useful operational state.

(37) Considering the warm-up or heat-up embodiments introduced above, the balance of this disclosure and terms used should be viewed in a light that also considers the design option for such warm-up or heat-up. Thus, where terms such as cooling channel, cooling chamber volume, cavity, and cooling port are used, for example, such terms could also be considered as a thermal control channel, a thermal control volume, or a thermal control port, respectively. A person of skill would understand that heat flux or heat transfer would go in a different direction, but the design concepts are similar and can be successfully employed in the various embodiments.

(38) In some embodiments, a cooling channel is a liquid cooling channel, and a liquid may flow through the liquid cooling channel. In some embodiments, the liquid may comprise a water and/or glycol (e.g.,

propylene glycol, ethylene glycol, and mixtures thereof).

(39) As described below, coolant fluid flowing through a cold plate may be used to control the temperature of semiconductor devices. The fluid flowing across the surface of the semiconductor device absorbs heat and conducts heat away from the semiconductor device.

(40) FIG. 1 is a schematic side view of a device package 10 and a heat sink 22 attached to the device package 10. The device package 10 typically includes a package substrate 12, a first device 14, a device stack 15, a heat spreader 18, and first TIM layers 16A, 16B thermally coupling the first device 14 and the device stack 15 to the heat spreader 18. The device package 10 is thermally coupled to the heat sink 22 through a second TIM layer 20. The TIM layers 16A, 16B, 20 facilitate thermal contact between components in the device package 10 and between the device package 10 and the heat sink 22.

(41) As heat flux density increases with increasing power density in advanced semiconductor devices, the cumulative thermal resistance of the system illustrated in FIG. 1 is increasingly problematic as heat cannot be dissipated quickly enough to allow semiconductor devices to run at optimal power. Consequently, the energy efficiency of semiconductor devices is reduced. Furthermore, heat is transferred between semiconductor devices within the device package 10, as shown with heat transfer path 24 (illustrated as a dashed line), where heat may be undesirably transferred from the first device 14 having a high heat flux, such as a central processing unit (CPU) or a graphical processing unit (GPU), to the device stack 15 having low heat flux, such as memory, through the heat spreader 18.

(42) For example, as shown in FIG. 1, each device package component and the respective interfacial boundaries therebetween have a corresponding thermal resistance that forms heat transfer path 26 (illustrated by arrow 26 in FIG. 1). The right-hand side of FIG. 1 illustrates the heat transfer path 26 as a series of thermal resistances R1-R8 between a heat source and a heat sink. Here, R1 is the thermal resistance of the bulk semiconductor material of the first device 14. R3 and R7 are the thermal resistances of the first TIM layers 16A, 16B and the second TIM layer 20, respectively. R5 is the thermal resistance of the heat spreader 18. R2, R4, R6, and R8 represent the thermal resistance at the interfacial region of the components (e.g., contact resistances). In a typical cooling system, R3 and R7 may account for 80% or more of the cumulative thermal resistance of the heat transfer path 26, and R5 may account for 5% or more. R1 of the first device 14 and R2, R4, R6, and R8 of the interfaces account for the remaining cumulative thermal resistance. Accordingly, embodiments described herein provide for integrated cooling assemblies embedded within a device package. The embedded cooling assemblies shorten the thermal resistance path between a semiconductor device and a heat sink and reduce thermal communication between semiconductor devices disposed in the same device package, such as described in relation to the figures below.

(43) FIG. 2A is a schematic plan view of an example of a system panel 100, in accordance with some embodiments of the present disclosure. Generally, the system panel 100 includes a printed circuit board (PCB) 102, a plurality of device packages 201 mounted to the PCB 102, and a plurality of coolant lines 108 fluidly coupling each of the device packages 201 to a coolant source 110. It is contemplated that coolant fluid may be delivered to each of the device packages 201 in any desired fluid phase (e.g., liquid, vapor, gas, or combinations thereof) and may flow out from each device package 201 in the same phase or a different phase. In some embodiments, the coolant fluid is delivered to the device packages 201 and returned therefrom as a liquid, whereby the coolant source 110 may comprise a heat exchanger or chiller to maintain the coolant fluid at a desired temperature. In other embodiments, the coolant fluid may be delivered to the device packages 201 as a liquid, vaporized to a vapor within the device packages 201, and returned to the coolant source 110 as a vapor. In those embodiments, the device packages 201 may be fluidly coupled to the coolant source 110 in parallel, and the coolant source 110 may include or further include a compressor (not shown) for condensing the received vapor to a liquid form.

(44) FIG. 2B is a schematic partial sectional side view of a portion of the system panel 100 of FIG. 2A, in accordance with some embodiments of the disclosure. As shown, each device package 201 is fluidly coupled to the plurality of coolant lines 108 and is disposed in a socket 114 of the PCB 102 and connected thereto using a plurality of pins 116, or by other suitable connection methods, such as solder bumps (not shown). The device package 201 may be seated in the socket 114 and secured to the PCB 102 using a mounting frame 106 and a plurality of fasteners 112, (e.g., compression screws, collectively configured to exert a relatively uniform downward force on the upward facing edges of the device package 201). The uniform downward force ensures proper pin contact between the device package 201 and the socket 114.

(45) FIG. 2C is a schematic exploded isometric view of an example device package 201, in accordance with

some embodiments of the present disclosure. Generally, the device package **201** includes a package substrate **202**, an integrated cooling assembly **203** disposed on the package substrate **202**, and a package cover **208** disposed on a peripheral portion of the package substrate **202**. Suitable materials that may be used in the package cover **208** include copper, aluminum, metal alloys, etc. The package cover **208** extends over the integrated cooling assembly **203** so that the integrated cooling assembly **203** is disposed between the package substrate **202** and the package cover **208**. The integrated cooling assembly **203** typically includes a semiconductor device **204** and a cold plate **206** bonded to the semiconductor device **204**. In some embodiments, the cold plate **206** may comprise substrate material like silicon, glass, ceramic, etc. In some embodiments, the lateral dimensions, or footprint, of the cold plate **206** is the same size or larger than the lateral dimensions, or footprint, of the semiconductor device **204**. In some embodiments, the footprint of the cold plate, as viewed from a top plan view, is the same size, or is larger than the footprint of the semiconductor device, as viewed from a top plan view. In some embodiments, the footprint of the cold plate **206** may be larger or the same size in one or both directions when compared to the footprint of the semiconductor device **204**.

(46) As shown, the device package **201** further includes a sealing material layer **222** that forms a coolant fluid impermeable barrier between the package cover **208** and the integrated cooling assembly **203** that prevents leaking of the coolant fluid outside of the cooling assembly and prevents coolant fluid from reaching an active side **218** (discussed below in relation to FIG. 3) of the semiconductor device **204** and causing damage thereto. In some embodiments, the sealing material layer **222** comprises an adhesive material that reliably attaches the package cover **208** to the integrated cooling assembly **203**. In some embodiments, the sealing material layer **222** comprises a polymer or epoxy material that extends upwardly from the package substrate **202** to encapsulate and/or surround at least a portion of the semiconductor device **204**. In some embodiments, the sealing material layer **222** may also comprise conductive material (e.g., solder). In other embodiments, the sealing material layer **222** is formed from a molding compound (e.g., a thermoset resin), that when polymerized, forms a hermetic seal between the package cover **208** and the cold plate **206**. Here, the coolant fluid is delivered to the cold plate **206** through openings **222A** disposed through the sealing material layer **222**. As shown, the openings **222A** are respectively in registration and fluid communication with inlet and outlet openings **212** of the package cover **208** thereabove and inlet and outlet openings **206A** in the cold plate **206** therebelow. In some embodiments, the inlet and outlet openings **206A** in the cold plate **206** are outside of the footprint of the semiconductor device.

(47) It will be understood that the openings are shown in a section view. The openings may have any cross-sectional shape that allows fluid to flow therethrough (e.g., rectangular, square, hexagonal or circular cross-sections). For example, the inlet and outlet openings **206A** of the cold plate **206** may form an elongated shape extending from one side of the cold plate **206** to another side of the cold plate **206**. For example, the inlet and outlet openings **206A** may form any shape having a length greater than a width in the X-Y plane (e.g., a rectangular or a trapezoidal shape). A shape in the X-Y plane of the openings **222A** disposed through the sealing material layer **222** may be substantially the same as the shape of the inlet and outlet openings **206A** of the cold plate **206** in the same place. Furthermore, it will be understood that all references to an opening throughout the present disclosure refer to an opening defined by a sidewall (e.g., opening sidewall).

(48) Generally, the package substrate **202** includes a rigid material, such as an epoxy or resin-based laminate, that supports the integrated cooling assembly **203** and the package cover **208**. The package substrate **202** may include conductive features disposed in or on the rigid material that electrically couples the integrated cooling assembly **203** to a system panel, such as the PCB **102**.

(49) FIG. 3 is a schematic sectional view in the X-Z plane of the device package **201** taken along line A-A' of FIG. 2C, in accordance with some embodiments of the disclosure. As illustrated in FIG. 3, the semiconductor device **204** includes the active side **218** that includes device components (e.g., transistors, resistors, and capacitors, formed thereon or therein), and a non-active side, here the semiconductor device backside **220**, opposite the active side **218**. As shown, the active side **218** is facing towards an interposer **254** and the package substrate **202**. In some embodiments, the interposer **254** is attached to the package substrate **202**. The active side **218** may be electrically connected to the package substrate **202** by use of conductive bumps **219**, which are encapsulated by a first underfill layer **221** disposed between the semiconductor device **204** and the package substrate **202**. The first underfill layer **221** may comprise a cured polymer resin or epoxy, which provides mechanical support to the conductive bumps **219** and protects against thermal fatigue. In some embodiments, the active side **218** may be electrically connected to another

package substrate, another active die, or another passive die (e.g., interposer **254**) using hybrid bonding or conductive bumps **219**. The cold plate **206** may be disposed above the package substrate **202** with the semiconductor device **204** disposed therebetween. For example, the semiconductor device **204** (and the first underfill layer **221**) may be disposed between the cold plate **206** and the package substrate **202**. In some embodiments, the cold plate **206** may be disposed directly on the package substrate **202** and/or on the interposer **254**.

(50) In some embodiments, the cold plate **206** has a first footprint and the semiconductor device **204** has a second footprint smaller than the first footprint. The device package **201** may comprise a first standoff **250** and a second standoff **252**. In some embodiments, the first standoff **250** and/or the second standoff **252** comprise a silicon, metal, underfill material, molding compound, and/or similar such material. The first standoff **250** and/or the second standoff **252** may be disposed between one or more portions of the cold plate **206** (that are located outside the second footprint of the semiconductor device **204**) and the interposer **254**. In some embodiments, the first standoff **250** and/or the second standoff **252** are adjacent to the semiconductor device **204** and provide support to the cold plate **206**. In some embodiments, the device package **201** comprises additional standoffs between the interposer **254** and portions of the cold plate **206** that are located outside the footprint of the semiconductor device **204**.

(51) Here, the cold plate **206** comprises a top portion **234** and a perimeter sidewall **240** (e.g., a perimeter sidewall defining a perimeter of the cold plate **206**) extending downwardly from the top portion **234** to level of the backside **220** of the semiconductor device **204**. In some embodiments, the top portion **234**, the perimeter sidewall **240**, and the backside **220** of the semiconductor device **204** collectively define a coolant channel **210** therebetween. In some embodiments, the cold plate **206** also comprises cavity dividers extending downwardly from the top portion **234** towards the backside **220** of the semiconductor device **204**. The cavity dividers may extend laterally and in parallel between an inlet opening **206A** of the cold plate **206** and an outlet opening **206A** of the cold plate **206** to define coolant channels **210** therebetween. In some embodiments, the cold plate **206** comprises one cavity divider which forms two coolant channels (e.g., one coolant channel on either side of the cavity divider) and portions of the perimeter sidewall **240**. In some embodiments, coolant channels **210** may be formed between a cavity divider and a portion of the perimeter sidewall **240** extending parallel to the cavity divider. Alternatively, in other embodiments, the cold plate **206** may comprise plural cavity dividers, for example two cavity dividers, five cavity dividers, or six cavity dividers. In such examples, the cold plate **206** comprises more than two coolant channels **210**, for example three coolant channels, four coolant channels, seven coolant channels, or more, defined between the cavity divider(s) and the perimeter sidewall **240**.

(52) In some embodiments, cavity dividers comprise cavity sidewalls which form surfaces of corresponding coolant channels **210**. In embodiments where plural cavity dividers extend in parallel to each other, cavity sidewalls of adjacent cavity dividers are opposite (e.g., facing) each other. In embodiments comprising a single cavity divider, a first cavity sidewall may be opposite (e.g., face) a first portion of the perimeter sidewall **240** extending parallel to and facing the first cavity sidewall. A second cavity sidewall may be opposite (e.g., face) a second portion of the perimeter sidewall **240** extending parallel to and facing the second cavity sidewall. The first portion of the perimeter sidewall **240** may be an opposite side of the cold plate **206** to the second portion of the perimeter sidewall **240**. For example, in embodiments where the cold plate **206** is rectangular, first and second opposing sides of the rectangular cold plate **206** form the first and second portions of the perimeter sidewall **240**.

(53) In some embodiments, cavity dividers may be continuous cavity dividers which extend continuously (e.g., in the Y-axis direction) between the inlet opening **206A** and the outlet opening **206A** of the cold plate **206**.

(54) In some embodiments, the backside **220** of the semiconductor device **204** comprises a corrosion protective layer (not shown). The corrosion protective layer may be a continuous layer disposed across the entire backside **220** of the semiconductor device **204**, such that the cold plate **206** is attached thereto. Beneficially, the corrosion protective layer provides a corrosion-resistant barrier layer, thus preventing undesired corrosion of the semiconductor device **204** (e.g., the semiconductor substrate material which might otherwise be in direct contact with coolant fluid flowing through a coolant channel **210**).

(55) One or more coolant chamber volumes may include one or more coolant channels. The coolant channels may extend between a single inlet opening and a single outlet opening of the cold plate **206**, such that the coolant chamber volume(s) and/or coolant channel(s) share the same inlet and outlet openings. In

some embodiments, multiple inlet and/or outlet openings may be coupled to the coolant chamber volume(s).

(56) In embodiments having plural coolant chamber volumes and/or plural coolant channels, each coolant chamber volume and/or coolant channel may be connected between a separate inlet opening and a separate outlet opening. In such embodiments, the coolant fluid may be directed to the separate inlet openings and from the separate outlet openings using a manifold disposed above the openings in the Z-axis direction.

(57) In some embodiments, a height in the Z-axis direction of the coolant chamber volume(s) and or coolant channel(s) may be greater than 100 μm , 100 μm -1000 μm , or 100 μm -700 μm . A width in the Y-axis direction of the coolant chamber volume(s) and/or coolant channel(s) may be greater than 100 μm , 100 μm -1000 μm , or 100 μm -700 μm . For example, the width of the coolant chamber volume(s) and/or coolant channel(s) may be greater than the height. A cross-section of the coolant chamber volume(s) and/or coolant channel(s) in the Y-Z plane is wide enough to allow for a pressure drop of 0-20 psi, 3-15 psi, or 4-10 psi.

(58) In some embodiments, preparing a desired surface roughness of the sidewalls of the coolant chamber volume(s) and/or coolant channels may include depositing an organic layer on a photoresist layer after cold plate features have been etched to form a micro-masking layer, such as between 1 to 30 nm. The micro-masking layer may be dry etched to form the desired surface roughness, such as between 0.1 to 3.0 nm.

(59) With reference to FIG. 3, the cold plate **206** is attached to the backside **220** of the device **204** without the use of an intervening adhesive. For example, the cold plate **206** may be directly bonded to the backside **220** of the device **204**, such that the cold plate **206** and the backside **220** of the device **204** are in direct contact. For example, in some embodiments, one or both of the cold plate **206** and the backside **220** of the semiconductor device **204** may comprise a dielectric material layer (e.g., a first dielectric material layer and a second dielectric material layer respectively), and the cold plate **206** is directly bonded to the backside **220** of the semiconductor device **204** through bonds formed between the dielectric material layers. In some embodiments, one of the cold plate **206** or the backside **220** of the semiconductor device **204** may comprise a thin bonding dielectric layer (e.g., silicon nitride, etc.) and other element(s) may not include any such explicit bonding dielectric layer (or can have only a native oxide layer). In some embodiments, one or more dielectric material layers are continuous layers. In some embodiments, one or more dielectric material layers may not be continuous. For example, a first dielectric material layer may be disposed only on lower surfaces of the cold plate **206** facing the backside **220** of the semiconductor device **204**. In some embodiments, portions of a first dielectric material layer may be disposed only on lower surfaces of support features **230** and the perimeter sidewall **240**. Beneficially, directly bonding the cold plate **206** to the semiconductor device **204**, as described above, reduces the thermal resistance therebetween and increases the efficiency of heat transfer from the semiconductor device **204** to the cold plate **206**. In some embodiments, thermal resistance is further reduced by directly bonding lower surfaces of cavity dividers facing the semiconductor device **204** to the backside **220** of the semiconductor device **204**.

(60) FIG. 4 is a schematic sectional view in the Y-Z plane of the integrated cooling assembly **203**, in accordance with some embodiments of the disclosure. In FIG. 4, the cold plate **206** comprises a patterned side that faces towards the semiconductor device **204** and an opposite side that faces towards the package cover **208** (not shown). The patterned side comprises a coolant chamber volume having plural coolant channels **210**, which extend laterally between the inlet and outlet openings **106A** of the cold plate **206**. Each coolant channel **210** comprises cavity sidewalls that define a corresponding coolant channel **210**. Portions of the cold plate **206** between the cavity sidewalls form support features **230**. The support features **230** provide structural support to the integrated cooling assembly **203** and disrupt laminar fluid flow at the interface of the coolant and the device backside **220**, resulting in increased heat transfer therebetween. Furthermore, by introducing plural coolant channels **210** to define separate coolant flow paths, an internal surface area of the cold plate **206** is increased, which further increases the efficiency of heat transfer.

(61) In FIG. 4, arrows **228A** and **228B** illustrate two different heat transfer paths in the integrated cooling assembly **203**. A first heat transfer path illustrated by arrow **228B** shows heat generated by the semiconductor device **204** transferring directly from the semiconductor material of the semiconductor device **204** to coolant fluid flowing through the cold plate **206**. A second heat transfer path illustrated by arrows **228A** shows heat generated by the semiconductor device **204** being transferred from semiconductor material (e.g., silicon material) of the semiconductor device **204** to semiconductor material (e.g., silicon material) of the cold plate **206** structure, propagated throughout the semiconductor material of the cold plate **206** structure (shown as dashed lines), and being transferring into coolant fluid flowing through the cold plate **206**. A thermal resistance of the first and second heat transfer paths **228A**, **228B** is illustrated by heat

transfer path **228C**, which is shown as thermal resistance **R1** between a heat source and a cold plate. Here, **R1** is the thermal resistance of the bulk semiconductor material of the semiconductor device **204**. It can be seen that the heat transfer path **228C** of the integrated cooling assembly **203** is reduced compared to the heat transfer path **26** of the device package **10** of FIG. **1**, due to the direct bonding discussed above.

(62) In some embodiments, the cold plate **206** may be attached to the semiconductor device **204** using a hybrid bonding technique, where bonds are formed between one or more dielectric material layers and between metal features, such as between first metal pads and second metal pads, disposed in the one or more dielectric material layers.

(63) Suitable dielectrics that may be used as the one or more dielectric material layers include silicon oxides, silicon nitrides, silicon oxynitrides, silicon carbon nitrides, metal-oxides, metal-nitrides, silicon carbide, silicon oxycarbides, silicon oxycarbonitride, diamond-like carbon (DLC), or combinations thereof. In some embodiments, one or more of the dielectric material layers are formed of an inorganic dielectric material (e.g., a dielectric material substantially free of organic polymers). Typically, one or more of the dielectric layers are deposited to a thickness greater than the thickness of a native oxide, such as about 1 nanometer (nm) or more, 5 nm or more, 10 nm or more, 50 nm or more, or 100 nm or more. In some embodiments, one or more of the dielectric layers are deposited to a thickness of 3 micrometers or less, 1 micrometer or less, 500 nm or less, such as 100 nm or less, or 50 nm or less. The dielectric layer material and thickness may be optimized for lower thermal resistance between the die and the cold plate.

(64) The cold plate **206** may be formed of any suitable material that has sufficient structural strength to withstand the desired pressures of coolant flowing into the coolant channels **210**. For example, the cold plate **206** may be formed of semiconductor material like silicon or other engineered materials like glass. In other examples, the cold plate **206** may be formed of a material selected from a group comprising polymers, metals, ceramics, or composites thereof. In some embodiments, the cold plate **206** may be formed of stainless steel (e.g., from a stainless-steel metal sheet) or a sapphire plate.

(65) In some embodiments, the cold plate **206** may be formed of a bulk material having a substantially similar coefficient of linear thermal expansion (CTE) to the bulk material of the substrate **202** and/or the semiconductor device **204**, where the CTE is a fractional change in length of the material (in the X-Y plane) per degree of temperature change. In some embodiments, the CTEs of the cold plate **206**, the substrate **202**, and/or the semiconductor device **204** are matched so that the CTE of the substrate **202** and/or the semiconductor device **204** is within about $\pm 20\%$ or less of the CTE of the cold plate **206**, such as within $\pm 15\%$ or less, within $\pm 10\%$ or less, or within about $\pm 5\%$ or less when measured across a desired temperature range. In some embodiments, the CTEs are matched across a temperature range from about -60°C . to about 100°C . or from about -60°C . to about 175°C . In one example embodiment, the matched CTE materials each include silicon.

(66) In some embodiments, the cold plate **206** may be formed of a material having a substantially different CTE from the semiconductor device **204** (e.g., a CTE mismatched material). In such embodiments, the cold plate **206** may be attached to the semiconductor device **204** by a compliant adhesive layer (not shown) or a molding material that absorbs the difference in expansion between the cold plate **206** and the semiconductor device **204** across repeated thermal cycles.

(67) The package cover **208** shown in FIGS. **2C** and **3** generally comprises one or more vertical or sloped sidewall portions **208A** and a lateral portion **208B** that spans and connects the sidewall portions **208A**. The sidewall portions **208A** may extend upwardly from a peripheral surface of the package substrate **202** to surround the device **204** and the cold plate **206** disposed thereon. The lateral portion **208B** may be disposed over the cold plate **206** and is typically spaced apart from the cold plate **206** by a gap corresponding to the thickness of the sealing material layer **222**. Coolant is circulated through the coolant chamber channels **210** through the inlet and outlet openings **212** of the package cover **208** formed through the lateral portion **208B**. The inlet and outlet openings **206A** of the cold plate **206** may be in fluid communication with the inlet and outlet openings **212** of the package cover **208** through the inlet and outlet openings **222A** formed in the sealing material layer **222** disposed therebetween. In certain embodiments, coolant lines **108** (FIGS. **2A-2B**) may be attached to the device package **201** by use of connector features formed in the package cover **208**, such as threads formed in the sidewalls of the inlet and outlet openings **212** of the package cover **208** and/or protruding features **214** that surround the inlet and outlet openings **212** and extend upwardly from a surface of the lateral portion **208B**.

(68) Typically, the package cover **208** is formed of semi-rigid or rigid material so that at least a portion of

the downward force exerted on the package cover **208** by the mounting frame is transferred to a supporting surface of the package substrate **202** and not transferred to the cold plate **206** and the semiconductor device **204** therebelow. In some embodiments, the package cover **208** is formed of a thermally conductive metal, such as aluminum or copper. In such embodiments, the package cover **208** functions as a heat spreader that redistributes heat from one or more electronic components of the semiconductor device **204**.

(69) It should be noted that the direction in which the coolant fluid flows through the cold plate **206** may be controlled depending on the relative locations of the inlet and outlet openings. For example, the coolant fluid may flow from left to right in the device package **201** of FIG. **3** when the inlet openings **212**, **222A**, **206A** of the package cover **208**, the sealing material layer **222**, and the cold plate **206**, respectively, are located on the left-hand side of the device package **201** and the outlet openings **212**, **222A**, **206A** of the package cover **208**, the sealing material layer **222**, and the cold plate **206**, respectively, are located on the right-hand side of the device package **201**. Alternatively, the coolant fluid may flow from right to left in the device package **201** illustrated in FIG. **3** when the outlet openings **212**, **222A**, **206A** of the package cover **208**, the sealing material layer **222**, and the cold plate **206** are located on the left-hand side of the device package **201** and the inlet openings **212**, **222A**, **206A** of the package cover **208**, the sealing material layer **222**, and the cold plate **206** are located on the right-hand side of the device package **201**. Although only one set of inlet and outlet openings is shown and described here, additional inlet and outlet openings may also be provided at various locations on the package cover **208**, the sealing material layer **222**, and the cold plate **206**.

(70) An example flow path of the coolant fluid through the coolant chamber volume may be as follows: 1. Coolant fluid enters the coolant channel **210** through the inlet openings. 2. Coolant fluid flows across the inside surfaces of the cold plate **206** and absorbs heat generated by the semiconductor device **204**, which has dissipated into the cold plate **206** structure. The coolant fluid may also flow directly across the backside **220** of the semiconductor device **204** to absorb heat energy directly from the semiconductor device **204**. The coolant chamber volume may additionally have various channels formed to direct the coolant fluid flow from inlet opening(s) to outlet opening(s) and facilitate heat extraction from the semiconductor device **204** by the coolant fluid. In some embodiments, the coolant fluid may be in direct contact with the backside **220** of the semiconductor device **204** or via one or more substrate or layers between the coolant fluid or backside **220** of the semiconductor device **204**. 3. Coolant fluid exits the coolant chamber volume **210** through outlet openings.

(71) It will be understood from the above flow path that heat is extracted without introducing an unnecessary thermal resistance (e.g., a TIM disposed between the backside **220** of the semiconductor device **204** and the cold plate **206**) between the backside **220** of the semiconductor device **204** and the cold plate **206**.

(72) FIG. **5** is a schematic side sectional view in the X-Z plane of an example of a multi-component device package **501** that includes a cold plate **506** directly bonded to the backside surfaces of two or more devices **501A**, **501B**. The multi-component device package **501** may be similar to the device package **201** described above, and therefore the description of similar features is omitted for brevity. In some embodiments, the two or more devices **501A** and **501B** are reconstituted and then bonded to the cold plate **506**. As shown, the device package **501** includes a package substrate **502**, an integrated cooling assembly **503** and a package cover **508**.

(73) The integrated cooling assembly **503** may include a plurality of devices **501A** (one shown), that may be singulated and/or disposed in a vertical device stack (e.g., device **501B**). The cold plate **506** may be attached to each of the devices **501A** and device stack **501B** (e.g., by the direct bonding methods described herein or other methods including flip chip bonding, etc.). In some embodiments, the device **501A** may comprise a processor and the device stack **501B** may comprise a plurality of memory devices. Here, the device **501A** and the device stack **501B** are disposed in a side-by-side arrangement on an interposer **554** attached to the package substrate **502**. The device **501A** and the device stack **501B** may be in electrical communication with one another through conductive elements formed in, on, or through the interposer **554**. Here, the cold plate **506** is sized to provide a bonding surface for attachment to both the device **501A** and the device stack **501B** but may otherwise be the same or substantially similar to other cold plates described herein.

(74) In some embodiments, the lateral dimensions (or footprint) of the cold plate **506** may be the same or larger than the combined lateral dimensions (or footprint) of both the device **501A** and the device stack **501B**. In some embodiments, one or more sidewalls of the cold plate **506** may extend past the vertical

sidewalls of the device **501A** and the device stack **501B**. In some embodiments, more than one cold plate **506** may be bonded. For example, separate cold plates may be bonded to the device **501A** and the device stack **501B**. In some embodiments, the footprint of the cold plate **506** may be larger or the same size in one or both directions when compared to the combined footprint of both the device **501A** and the device stack **501B**. In some embodiments, the inlet and outlet openings in the cold plate are outside of the footprints of both the device **501A** and the device stack **501B**.

(75) In some embodiments, the cold plate **506** has a first footprint and the device **501A** and the device stack **501B** have a combined second footprint smaller than the first footprint. The device package **501** may comprise a first standoff **550**, a second standoff **556**, and a third standoff **552**. In some embodiments, the first standoff **550**, the second standoff **556**, and the third standoff **552** comprise a silicon, metal, underfill material, molding compound, and/or similar such material. The first standoff **550** and/or the third standoff **552** may be disposed between one or more portions of the cold plate **506** (that are located outside the combined second footprint of the device **501A** and the device stack **501B**) and the interposer **554**. In some embodiments, the first standoff **550** is adjacent to the device **501A** and provides support to the cold plate **506**. In some embodiments, the third standoff **552** is adjacent to the device stack **501B** and provides support to the cold plate **506**.

(76) In some embodiments, there may be a gap between the device **501A** and the device stack **501B**. The second standoff **556** may be disposed within the gap. For example, the second standoff **556** may be adjacent to the device **501A** and the device stack **501B**. The second standoff **556** may also be between the cold plate **506** and the interposer **554** to provide support to the cold plate **506**. Although three standoffs are shown, the device package **501** may comprise any number of standoffs. For example the device package **501** may comprise one, two, five, or ten standoffs.

(77) FIG. **6** is a flow diagram showing a method **60** of forming an integrated cooling assembly, according to embodiments of the present disclosure. Generally, the method **60** includes bonding a first substrate comprising one or more cold plates **206** to a second substrate comprising one or more semiconductor devices **204**, and singulating one or more integrated cooling assemblies **203** from the bonded first and second substrates. For example, a wafer (bare or reconstituted wafer) comprising one or more cold plates **206** can be directly bonded to another wafer (bare or reconstituted wafer) comprising one or more semiconductor devices **204**.

(78) It will be understood that the first substrate may be a cold plate die or part of a wafer of cold plates. Further, the second substrate may be a semiconductor device die or part of a wafer of semiconductor devices **204**. Therefore, the method **60** may include die-to-die direct bonding (e.g., cold plate die to semiconductor device die), wafer-to-die direct bonding (e.g., cold plate die to semiconductor device wafer, or cold plate wafer to semiconductor device die), and wafer-to-wafer direct bonding (e.g., cold plate wafer to semiconductor device wafer). It will be understood that the singulation step (discussed in relation to block **64**, below) may not be required for a die-to-die direct bonding operation.

(79) For simplicity, the following description is focused on forming one integrated cooling assembly **203** comprising one cold plate **206** and one semiconductor device **204**. However, as mentioned above, in some embodiments, the first substrate may comprise plural cold plates **206** and the second substrate may comprise plural semiconductor devices **204**, such that plural integrated cooling assemblies **203** may be formed from the first and second substrates.

(80) At block **62**, the method **60** includes directly bonding the first substrate (e.g., a monocrystalline silicon wafer) comprising a cold plate **206** to the second substrate (e.g., a monocrystalline silicon wafer) comprising a semiconductor device **204** without an intervening adhesive. In some embodiments, the cold plate **206** may be directly bonded to the backside of the semiconductor device **204**.

(81) In some embodiments, the first substrate may be etched using a patterned mask layer formed on its surface to form features of the cold plate **206**. An anisotropic etch process may be used, which uses inherently differing etch rates for the silicon material as between {100} plane surfaces and {111} plane surfaces when exposed to an anisotropic etchant.

(82) In some embodiments, the etching process is controlled to where the etch rates of the substrate surfaces have a ratio between about 1:10 and about 1:200, such as between about 1:10 and about 1:100, for example between about 1:10 and 1:50, or between about 1:25 and 1:75. Examples of suitable anisotropic wet etchants include aqueous solutions of potassium hydroxide (KOH), ethylene diamine and pyrocatechol (EPD), ammonium hydroxide (HN.sub.4OH), hydrazine (N.sub.2H.sub.4), or tetra methyl ammonium hydroxide

(TMAH). The actual etch rates of the silicon substrate depend on the concentration of the etchant in the aqueous solution, the temperature of the aqueous solution, and a concentration of the dopant in the substrate (if any). Typically, the mask layer is formed of a material that is selective to anisotropic etch compared to the underlying monocrystalline silicon substrate. Examples of suitable mask materials include silicon oxide (Si.sub.xO.sub.y) or silicon nitride (Si.sub.xN.sub.y). In some embodiments, the mask layer has a thickness of about 100 nm or less, such as about 50 nm or less, or about 30 nm or less. The mask layer may be patterned using any suitable combination of lithography and material etching patterning methods.

(83) The second substrate may include a bulk material, and a plurality of material layers disposed on the bulk material. The bulk material may include any semiconductor material suitable for manufacturing semiconductor devices, such as silicon, silicon carbide, silicon germanium, germanium, group III-V semiconductor materials, group II-VI semiconductor materials, or combinations thereof. While some high-performance processors like CPUs, GPUs, neural processing units (NPU), and tensor processing units (TPUs) are typically made out of silicon, some other high-power density (hence substantial heat-generating) devices may comprise silicon carbide or gallium nitride, for example. In some embodiments, the second substrate may include a monocrystalline wafer, such as a silicon wafer, a plurality of device components formed in or on the silicon wafer, and a plurality of interconnect layers formed over the plurality of device components. In other embodiments, the second substrate may comprise a reconstituted substrate (e.g., a substrate formed from a plurality of singulated devices embedded in a support material). In some embodiments, each semiconductor device may have its own individual cold plate fabricated through a reconstitution process.

(84) The bulk material of the second substrate may be thinned after the semiconductor device **204** is formed using one or more backgrinding, etching, and polishing operations that remove material from the backside. Thinning the second substrate may include using a combination of grinding and etching processes to reduce the thickness (in the Z-direction) to about 450 μm or less, such as about 200 μm or less, or about 150 μm or less or about 50 μm or less. After thinning, the backside **220** may be polished to a desired smoothness using a chemical mechanical polishing (CMP) process, and the dielectric material layer may be deposited thereon. In some embodiments, the dielectric material layer may be polished to a desired smoothness to prepare the second substrate for the bonding process. In some embodiments, the method **60** includes forming a plurality of metal features in the dielectric material layer in preparation for a hybrid bonding process, such as by use of a damascene process.

(85) In some embodiments, the active side of the second substrate is temporarily bonded to a carrier substrate (not shown) before or after the thinning process. When used, the carrier substrate provides support for the thinning operation and/or for the thinned material to facilitate substrate handling during one or more of the subsequent manufacturing operations described herein.

(86) Here, the method **60** may include forming dielectric layers on one or both the first and second substrates, and directly bonding includes forming dielectric bonds between a first dielectric material layer of the first substrate and a second dielectric material layer of the second substrate (or forming dielectric bonds between one substrate and a dielectric material layer of the other substrate). Direct bonding processes join dielectric layers by forming strong chemical bonds (e.g., covalent bonds) between the dielectric layers.

(87) Generally, directly bonding the surfaces (of the dielectric material layers formed on the first and second substrates) includes preparing, aligning, and contacting the surfaces. Examples of dielectric material layers include silicon oxide, silicon nitride, silicon oxynitride, and silicon carbonitride. Preparing the surfaces may include smoothing the respective surfaces to a desired surface roughness, such as between 0.1 to 3.0 nm RMS, activating the surfaces to weaken or open chemical bonds in the dielectric material, and terminating the surfaces with a desired species. Smoothing the surfaces may include polishing the first and second substrates using a CMP process. Activating and terminating the surfaces with a desired species may include exposing the surfaces to radical species formed in a plasma. The bond interface between the bonded dielectric layers can include a higher concentration of materials from the activation and/or last chemical treatment processes compared to the bulk of the bonding layers. For example, in some embodiments that utilize a nitrogen plasma for activation, a nitrogen concentration peak can be formed at the bond interface. In some embodiments, the nitrogen concentration peak may be detectable using secondary ion mass spectroscopy (SIMS) techniques. In various embodiments, for example, a nitrogen termination treatment (e.g., exposing the bonding surface to a nitrogen-containing plasma) can replace OH groups of a hydrolyzed (OH-terminated) surface with NH.sub.2 molecules, yielding a nitrogen-terminated surface. In embodiments

that utilize an oxygen plasma for activation, an oxygen concentration peak can be formed at the bond interface between non-conductive bonding surfaces.

(88) In some embodiments, the plasma is formed using a nitrogen-containing gas (e.g., N₂), and the terminating species includes nitrogen, or nitrogen and hydrogen. In some embodiments, fluorine may also be present within the plasma. In some embodiments, the surfaces may be activated using a wet cleaning process (e.g., by exposing the surfaces to an aqueous ammonia solution). In some embodiments, the dielectric bonds may be formed using a dielectric material layer deposited on only one of the first and second substrates, but not on both. In those embodiments, the direct dielectric bonds may be formed by contacting the deposited dielectric material layer of one of the first and second substrates directly with a bulk material surface (or such a surface with a native oxide) of the other substrate.

(89) Directly forming direct dielectric bonds between the first and second substrates at block **62** may include bringing the prepared and aligned surfaces into direct contact at a temperature less than 150° C., such as less than 100° C., for example, less than 30° C., or about room temperature, e.g., between 20° C. and 30° C. Without intending to be bound by theory, in the case of directly bonding surfaces terminated with nitrogen and hydrogen (e.g., NH₂ groups), it is believed that the hydrogen terminating species diffuse from the interfacial bonding surfaces, and chemical bonds are formed between the remaining nitrogen species during the direct bonding process. In some embodiments, the direct bond is strengthened using an anneal process, where the substrates are heated to and maintained at a temperature of greater than about 30° C. and less than about 450° C., for example, greater than about 50° C. and less than about 250° C., or about 150° C., for a duration of about 5 minutes or more, such as about 15 minutes. Typically, the bonds will strengthen over time even without the application of heat. Thus, in some embodiments, the method does not include heating the substrates.

(90) In embodiments where the first and second substrates are bonded using hybrid dielectric and metal bonds, the method **60** may further include planarizing or recessing the metal features below the dielectric field surface before contacting and bonding the dielectric material layers. After the dielectric bonds are formed, the first and second substrates may be heated to a temperature of 150° C. or more and maintained at the elevated temperature for a duration of about 1 hour or more (e.g., between 8 and 24 hours) to form direct metallurgical bonds between the metal features.

(91) Suitable direct dielectric and hybrid bonding technologies that may be used to perform aspects of the methods described herein include ZiBond® and DBI®, each of which are commercially available from Adeia Holding Corp., San Jose, CA, USA.

(92) In some embodiments, one or more standoffs are attached to the second substrate prior to the first substrate comprising the cold plate **206** being directly bonded to the second substrate comprising the semiconductor device **204**. The one or more standoffs may be attached using one or more adhesives. The one or more standoffs may be disposed adjacent to the semiconductor device **204**. For example, the first standoff **250** may be attached to the substrate adjacent to a first side of the semiconductor device **204** and the second standoff **252** may be attached to the substrate adjacent to a second side of the semiconductor device **204**.

(93) In some embodiments, one or more standoffs are attached to the cold plate **206** prior to the first substrate comprising the cold plate **206** being directly bonded to the second substrate comprising the semiconductor device **204**. The one or more standoffs may be attached using one or more adhesives or direct bonding. For example, the first standoff **250** may be directly bonded to a first portion of the cold plate **206** and the second standoff **252** may be directly bonded to a second portion of the cold plate **206**.

(94) In some embodiments, one or more portions of a manifold are attached to the second substrate prior to the first substrate comprising the cold plate **206** being directly bonded to the second substrate comprising the semiconductor device **204**. The one or more portions of the manifold may be attached using one or more adhesives. The one or more portions of the manifold may be disposed adjacent to the semiconductor device **204**. For example, a first portion **1450** of a manifold **1460** may be attached to the substrate adjacent to a first side of the semiconductor device **204** and a second portion **1452** of the manifold **1460** may be attached to the substrate adjacent to a second side of the semiconductor device **204**.

(95) In some embodiments, one or more portions of the manifold are attached to the cold plate **206** prior to the first substrate comprising the cold plate **206** being directly bonded to the second substrate comprising the semiconductor device **204**. The one or more standoffs may be attached using one or more adhesives. For example, the first portions **1450** of the manifold **1460** may be attached to a first portion of the cold plate **206**

and the second portion **1452** of the manifold **1460** may be directly bonded to a second portion of the cold plate **206**.

(96) At block **64**, the method **60** includes singulating at least one integrated cooling assembly **203** from the bonded first and second substrates. Singulation after bonding may impart distinctive structural characteristics on the integrated cooling assembly **203** as the bonding surface of the cold plate **206** may have a different perimeter than the backside of the semiconductor device **204** bonded thereto. In some embodiments, the sidewalls (e.g., side surfaces) of the cold plate **206** may extend past the edges (e.g., side surfaces) of the semiconductor device **204**. In some embodiments, the cold plate **206** is singulated from the first substrate using a process that cuts or divides the first substrate in a vertical plane (i.e., in the Z-direction). In those embodiments, the side surfaces of the cold plate **206** are substantially perpendicular to the backside **220** of the semiconductor device **204** (i.e., a horizontal (X-Y) plane of an attachment interface between the semiconductor device **204** and the cold plate **206**). In some embodiments, the cold plate **206** is singulated using a saw or laser dicing process.

(97) At block **66**, the method **60** may include connecting the integrated cooling assembly **203** to the package substrate **202** and sealing a package cover **208** comprising inlet and outlet openings **212** to the integrated cooling assembly **203** by use of a molding compound that, when cured, forms a sealing material layer **222**.

(98) At block **68**, the method **60** may include, before or after sealing the package cover **208** to the integrated cooling assembly **203**, forming inlet and outlet openings **222A** in the sealing material layer **222** to fluidly connect the inlet and outlet openings **212** of the package cover **208** to the cold plate **206**.

(99) FIG. **7** shows a schematic top view of a portion of a device package **701** that may be used with a system panel, in accordance with some embodiments of the present disclosure. The portion of the device package **701** may be the same or similar to portions of the device package **201** described above, and therefore the description of similar features is omitted for brevity. In some embodiments, the portion of the device package **701** comprises a cold plate **702** bonded to a device **704**. Only the outline of the device **704** is shown in FIG. **7** because the device **704** is located under the cold plate **702**. In some embodiments, the cold plate **702** comprises a first region, a second region, and a third region, wherein the second region is between the first region and the third region. In some embodiments, the cold plate **702** is bonded to the device **704** via the first region and the third region.

(100) In some embodiments, the cold plate **702** comprises an inlet **706**, an outlet **708**, and a plurality of coolant channels (e.g., a first coolant channel **710**). Although six coolant channels are shown, the cold plate **702** may comprise more or fewer coolant channels. Similarly, although one inlet **706** and one outlet **708** are shown, the cold plate **702** may comprise more or fewer inlets and/or outlets. In some embodiments, one or more coolant channels of the plurality of coolant channels (e.g., the first coolant channel **710**) extend along the second region of the cold plate **702**.

(101) In some embodiments, the lateral dimensions (or footprint) of the cold plate **702** are larger than the lateral dimensions (or footprint) of the device **704**. For example, one or more edges of the cold plate **702** may extend past the edges of the device **704**. In some embodiments, the cold plate **702** having a larger footprint than the footprint of the device **704** allows for more coolant to flow across more of the surface area of the device **704** resulting in more heat being dissipated from the device **704**.

(102) In some embodiments, the inlet **706** and outlet **708** are outside of the footprint of the device **704**. The inlet **706** and outlet **708** allow coolant to flow through the plurality of coolant channels (e.g., the first coolant channel **710**). In some embodiments, having the inlet **706** and outlet **708** of the cold plate **702** located outside the footprint of the device **704** allows for a reduction in stagnant coolant flows forming around the inlet **706** and outlet **708** regions, resulting in increased heat dissipation and decreased temperatures for the device **704**.

(103) FIG. **8** is a schematic sectional view in the X-Z plane of the portion of the device package **701** taken along line B-B' of FIG. **7**. The portion of the device package **701** may be the same or similar to portions of the device package **201** described above, and therefore the description of similar features is omitted for brevity. As described above, the portion of the device package **701** comprises the cold plate **702** bonded to the device **704**. The cold plate **702** comprises an inlet (not shown in FIG. **8**), an outlet (not shown in FIG. **8**), and a plurality of coolant channels (e.g., first coolant channel **710**).

(104) In some embodiments, a first portion **712** of the cold plate **702** and a second portion **714** of the cold plate **702** extend past one or more edges of the device **704** in the X-axis direction. In some embodiments, the first portion **712** and/or the second portion **714** of the cold plate **702** extending past one or more edges of

the device **704** in the X-axis direction allow for increased heat dissipation.

(105) FIG. **9** is a schematic sectional view in the Y-Z plane of the portion of the device package **701** taken along line C-C' of FIG. **7**. The portion of the device package **701** may be the same or similar to portions of the device package **201** described above, and therefore the description of similar features is omitted for brevity. As described above, the portion of the device package **701** comprises the cold plate **702** bonded to the device **704**. The cold plate **702** comprises the inlet **706**, the outlet **708**, and the plurality of coolant channels (e.g., first coolant channel **710**).

(106) In some embodiments, a third portion **716** of the cold plate **702** and a fourth portion **718** of the cold plate **702** extend past one or more edges of the device **704** in the Y-axis direction. In some embodiments, the third portion **716** and/or the fourth portion **718** of the cold plate **702** extending past one or more edges of the device **704** in the Y-axis direction allow for increased heat dissipation.

(107) As shown, the inlet **706** and outlet **708** are outside of the footprint of the device **704** in the Y-axis direction. The inlet **706** and outlet **708** allow coolant to flow through the plurality of coolant channels (e.g., the first coolant channel **710**). For example, coolant may flow from one or more coolant lines (e.g., coolant line **108**) into the inlet **706**. The coolant may then flow from the inlet **706** into the first coolant channel **710**. The coolant may then flow from the first coolant channel **710** to the outlet **708** and exit the cold plate **702**. In some embodiments, coolant flowing through the first coolant channel **710** directly contacts the backside of the device **704**. In some embodiments, the direct contact between coolant flowing through the first coolant channel **710** and the backside of the device **704** results in increased thermal dissipation and a decrease in temperature of the device **704**.

(108) In some embodiments, having the inlet **706** and outlet **708** of the cold plate **702** located outside the footprint of the device **704** allows for a reduction in stagnant coolant flows forming around the inlet region (e.g., third portion **716** of the cold plate **702**) and/or outlet region (e.g., fourth portion **718** of the cold plate **702**), resulting in increased heat dissipation and decreased temperatures for the device **704**.

(109) In some embodiments, the length of one or more coolant channels (e.g., first coolant channel **710**) that extends along the backside of the device **704** in the Y-axis direction is less than the length of the backside of the device **704** in the Y-axis direction. For example, the first coolant channel **710** may have a first length **720** in the Y-axis direction and the backside of the device **704** may have a second length **722** in the Y-axis direction. The difference between the first length **720** and the second length **722** may be the amount of cold plate material needed to provide adequate adhesion between the cold plate **702** and the device **704**. For example, the cold plate **702** may be attached to the device **704** by bonding a first portion of cold plate material **724** and a second portion of cold plate material **726** with the device **704**. In some embodiments, the first length **720** may be between 50 micrometers to 1 millimeter less than the second length **722** to provide adequate adhesion between the cold plate **702** and the device **704**.

(110) In some embodiments, the portion of the device package **701** comprises a first standoff **750** and a second standoff **752**. The first standoff **750** may be attached to the third portion **716** of the cold plate **702** and the second standoff **752** may be attached to the fourth portion **718** of the cold plate **702**. In some embodiments, the first standoff **750** and/or the second standoff **752** are adjacent to the device **704** and provide support to the cold plate **702**. In some embodiments, the portion of the device package **701** comprises additional or fewer standoffs than shown.

(111) FIGS. **10-11** show schematic top views of portions of device packages that were used in experimentations to test one or more aspects of the invention described herein. FIG. **10** displays a cold plate **1002**, a plurality of resistors (e.g., a first resistor **1006**), and a substrate **1004**. In some embodiments, the plurality of resistors are bonded to the substrate **1004** and the cold plate **1002** is attached to the plurality of resistors. As shown, the footprint of the cold plate **1002** is smaller than the footprint of the plurality of resistors. For example, a first portion **1008** of the plurality of resistors and a second portion **1010** of the plurality of resistors extend past one or more edges of the cold plate **1002** in the X-axis direction and a third portion **1012** of the plurality of resistors extend past one or more edges of the cold plate **1002** in the Y-axis direction. Temperature measurements were recorded during operations of the portion of the device package displayed in FIG. **10**. Temperature measurements recorded for the first portion **1008**, second portion **1010**, and third portion **1012** of the plurality of resistors were much higher than temperature measurements recorded for portions of the plurality of resistors located within the footprint of the cold plate **1002**.

(112) FIG. **11** displays a cold plate **1102**, a plurality of resistors (e.g., a first resistor **1106**), and a substrate **1104**. In some embodiments, the plurality of resistors are bonded to the substrate **1104** and the cold plate

1102 is attached to the plurality of resistors. As shown, the footprint of the cold plate **1102** is the same or similar to the footprint of the plurality of resistors in the X-axis direction but is smaller than the footprint of the plurality of resistors in the Y-axis direction. For example, a first portion **1112** of the plurality of resistors extends past one or more edges of the cold plate **1102** in the Y-axis direction. Temperature measurements were recorded during operations of the portion of the device package displayed in FIG. **11**. The portion of the device package displayed in FIG. **11** experienced overall decreased temperatures compared to the portion of the device package displayed in FIG. **10**. This result is caused, at least in part, on less portions of the plurality of resistors extending past the footprint of the cold plate **1102**.

(113) FIG. **12** shows a schematic sectional view of a portion of a device package **1201** that may be used with the system panel, in accordance with some embodiments of the present disclosure. The portion of the device package **1201** may be the same or similar to portions of device packages described above, and therefore the description of similar features is omitted for brevity. In some embodiments, the portion of the device package **1201** comprises a cold plate **1202** bonded to a device **1204**. The cold plate **1202** comprises an inlet **1206**, an outlet **1208**, a base plate **1210**, and one or more coolant channels (e.g., a first coolant channel **1212**).

(114) The device **1204** may have an active side electrically connected to a package substrate **1254** by use of conductive bumps **1256**, which are encapsulated by an underfill layer **1258** disposed between the semiconductor device **204** and the interposer **1214**. The underfill layer **1258** may comprise a cured polymer resin or epoxy, which provides mechanical support to the conductive bumps **1256** and protects against thermal fatigue. In some embodiments, the device **1204** may be electrically connected to another package substrate, another active die, or another passive die using hybrid bonding or conductive bumps **1256**. The cold plate **1202** may be disposed above the package substrate **1254** with the device **1204** disposed therebetween. For example, the device **1204**, the underfill layer **1258**, and the interposer **1214** may be disposed between the cold plate **1202** and the package substrate **1254**. In some embodiments, the package substrate **1254** has a larger footprint than the device **1204** and/or the cold plate **1202**.

(115) In some embodiments, the portion of the device package **1201** comprises a first standoff **1250** and a second standoff **1252**. In some embodiments, the first standoff **1250** and/or the second standoff **1252** comprise a silicon, metal, underfill material, molding compound, and/or similar such material. The first standoff **1250** and/or the second standoff **1252** may be disposed between the base plate **1210** and the package substrate **1254**. In some embodiments, the first standoff **1250** and/or the second standoff **1252** are adjacent to the device **1204**, the underfill layer **1258**, and/or the interposer **1214** and provide support to the cold plate **1202**. In some embodiments, the device package **1201** comprises additional or fewer standoffs than shown.

(116) In some embodiments, the portion of the device package **1201** comprises a manifold **1260** attached to the top of the cold plate **1202**. In some embodiments, the manifold **1260** comprises one or more channels in fluid communication with one or more coolant channels of the cold plate **1202**. For example, the first coolant channel **1212** of the cold plate **1202** may be in fluid communication with a second coolant channel **1262** and a third coolant channel **1264** of the manifold **1260**. In some embodiments, the manifold **1260** comprises one or more outlets and/or inlets. For example, the manifold **1260** may comprise a first inlet **1266** and a first outlet **1268**. In some embodiments, the first inlet **1266** is coupled to one or more coolant lines and allows coolant to flow from the coolant line to the cold plate **1202** (e.g., via the second coolant channel **1262**). In some embodiments, the first outlet **1268** is coupled to one or more coolant lines and allows coolant to flow from the cold plate **1202** to the one or more coolant lines (e.g., via the third coolant channel **1264**).

(117) FIG. **13** shows a schematic sectional view of a portion of a device package **1301** that may be used with the system panel, in accordance with some embodiments of the present disclosure. In some embodiments, the device package **1301** is the same or similar to the device package **1201** shown in FIG. **12**.

(118) In some embodiments, the device package **1301** comprises a first compression member **1302**, a second compression member **1304**, a third compression member **1306**, and a fourth compression member **1308**. In some embodiments, one or more compression members are compressed O-rings or gaskets. One or more compression members may experience a compression force. For example, a first clamp may be used to seal the manifold **1260** to the cold plate **1202**. The clamp may exert a compression force on the first compression member **1302**, the second compression member **1304**, the third compression member **1306**, and/or the fourth compression member **1308**. The first standoff **1250** and/or the second standoff **1252** may support the

cold plate 1202 during compression to prevent one or more portions of the cold plate 1202 from breaking. In some embodiments, more or less compression members may be used.

(119) FIG. 14 shows a schematic sectional view of a portion of a device package 1401 that may be used with the system panel, in accordance with some embodiments of the present disclosure. The portion of the device package 1401 may be the same or similar to portions of device packages described above, and therefore the description of similar features is omitted for brevity. In some embodiments, the portion of the device package 1401 comprises a cold plate 1402 bonded to a device 1404. The cold plate 1402 comprises an inlet 1406, an outlet 1408, a base plate 1410, and one or more coolant channels (e.g., a first coolant channel 1412).

(120) The device 1404 may have an active side electrically connected to a package substrate 1454 by use of conductive bumps 1456, which are encapsulated by an underfill layer 1458 disposed between the semiconductor device 1404 and the interposer 1414. The underfill layer 1458 may comprise a cured polymer resin or epoxy, which provides mechanical support to the conductive bumps 1456 and protects against thermal fatigue. In some embodiments, the device 1404 may be electrically connected to another package substrate, another active die, or another passive die using hybrid bonding or conductive bumps 1456. The cold plate 1402 may be disposed above the package substrate 1454 with the device 1404 disposed therebetween. For example, the device 1404, the underfill layer 1458, and/or the interposer 1414 may be disposed between the cold plate 1402 and the package substrate 1454. In some embodiments, the package substrate 1454 has a larger footprint than the device 1404 and/or the cold plate 1402.

(121) In some embodiments, the portion of the device package 1401 comprises a manifold 1460. In some embodiments, the manifold 1460 comprises one or more channels in fluid communication with one or more coolant channels of the cold plate 1402. For example, the first coolant channel 1412 of the cold plate 1402 may be in fluid communication with a second coolant channel 1462 and a third coolant channel 1464 of the manifold 1460. In some embodiments, the manifold 1460 comprises one or more outlets and/or inlets. For example, the manifold 1460 may comprise a first inlet 1466 and a first outlet 1468. In some embodiments, the first inlet 1466 is coupled to one or more coolant lines and allows coolant to flow from the coolant line to the cold plate 1402 (e.g., via the second coolant channel 1462). In some embodiments, the first outlet 1468 is coupled to one or more coolant lines and allows coolant to flow from the cold plate 1402 to the one or more coolant lines (e.g., via the third coolant channel 1464).

(122) In some embodiments, the manifold 1460 comprises a first portion 1450 and a second portion 1452. In some embodiments, the first portion 1450 and/or the second portion 1452 are disposed between the base plate 1410 and the package substrate 1454. In some embodiments, the first portion 1450 and/or the second portion 1452 are adjacent to the device 1404, the underfill layer 1458, and/or the interposer 1414 and provide support to the cold plate 1402.

(123) FIG. 15 shows a schematic sectional view of a portion of a device package 1501 that may be used with the system panel, in accordance with some embodiments of the present disclosure. In some embodiments, the device package 1501 is the same or similar to the device package 1401 shown in FIG. 14.

(124) In some embodiments, the device package 1501 comprises a first compression member 1502, a second compression member 1504, a third compression member 1506, and a fourth compression member 1508. In some embodiments, one or more compression members are compressed O-rings or gaskets. One or more compression members may experience a compression force. For example, a first clamp may be used to seal the manifold 1460 to the cold plate 1402. The clamp may exert a compression force on the first compression member 1502, the second compression member 1504, the third compression member 1506, and/or the fourth compression member 1508. The first portion 1450 and/or the second portion 1452 of the manifold 1460 may support the cold plate 1402 during compression to prevent one or more portions of the cold plate 1402 from breaking. In some embodiments, more or less compression member may be used.

(125) FIG. 16 shows a schematic sectional view of a portion of a device package 1601 that may be used with the system panel, in accordance with some embodiments of the present disclosure. The portion of the device package 1601 may be the same or similar to portions of device packages described above, and therefore the description of similar features is omitted for brevity. In some embodiments, the portion of the device package 1601 comprises a cold plate 1602 bonded to a device 1604.

(126) The device 1604 may have an active side electrically connected to a package substrate 1606 by use of conductive bumps 1608, which are encapsulated by a first underfill layer 1610 disposed between the device 1604 and the interposer 1612. The first underfill layer 1610 may comprise a cured polymer resin or epoxy,

which provides mechanical support to the conductive bumps **1608** and protects against thermal fatigue. In some embodiments, the first underfill layer **1610** comprises one or more edges. The one or more edges may comprise a sloped profile. For example, a first edge **1614** of the first underfill layer **1610** may have a sloped profile.

(127) In some embodiments, the device **1604** may be electrically connected to another package substrate, another active die, or another passive die using hybrid bonding or conductive bumps **1608**. The cold plate **1602** may be disposed above the package substrate **1606** with the device **1604** disposed therebetween. In some embodiments the device **1604**, the first underfill layer **1610**, and the interposer **1612** may be disposed between the cold plate **1602** and the package substrate **1606**. In some embodiments, the package substrate **1606** has a larger footprint than the device **1604** and/or the cold plate **1602**.

(128) In some embodiments, the portion of the device package **1601** comprises a second underfill layer **1616**. In some embodiments, the second underfill layer **1616** comprises a silicon, metal, underfill material, molding compound, and/or similar such material. One or more portions of the second underfill layer **1616** may be disposed between the cold plate **1602** and the package substrate **1606**. In some embodiments, one or more portions of the second underfill layer **1616** are adjacent to the device **1604**, the first underfill layer **1610**, and/or the interposer **1612** and provide support to the cold plate **1602**. In some embodiments, the second underfill layer **1616** comprises one or more edges. The one or more edges may comprise a sloped profile. For example, a first edge **1618** of the second underfill layer **1616** may have a sloped profile. In some embodiments, the sloped profile of one or more edges of the first underfill layer **1610** is the same or similar to the sloped profile of one or more edges of the second underfill layer **1616**.

(129) FIG. **17** shows a schematic sectional view of a portion of a device package **1701** that may be used with the system panel, in accordance with some embodiments of the present disclosure. The portion of the device package **1701** may be the same or similar to portions of device packages described above, and therefore the description of similar features is omitted for brevity. In some embodiments, the portion of the device package **1701** comprises a cold plate **1702** bonded to a device **1704**.

(130) The device **1704** may have an active side electrically connected to a package substrate **1706** by use of conductive bumps **1708**, which are encapsulated by a first underfill layer **1710** disposed between the device **1704** and the interposer **1712**. The first underfill layer **1710** may comprise a cured polymer resin or epoxy, which provides mechanical support to the conductive bumps **1708** and protects against thermal fatigue. In some embodiments, the first underfill layer **1710** comprises one or more edges. The one or more edges may comprise a sloped profile. For example, a first edge **1714** of the first underfill layer **1710** may have a sloped profile.

(131) In some embodiments, the device **1704** may be electrically connected to another package substrate, another active die, or another passive die using hybrid bonding or conductive bumps **1708**. The cold plate **1702** may be disposed above the package substrate **1706** with the device **1704** disposed therebetween. In some embodiments the device **1704**, the first underfill layer **1710**, and the interposer **1712** may be disposed between the cold plate **1702** and the package substrate **1706**. In some embodiments, the package substrate **1706** has a larger footprint than the device **1704** and/or the cold plate **1702**.

(132) In some embodiments, the portion of the device package **1701** comprises a second underfill layer **1716**. In some embodiments, the second underfill layer **1716** comprises a silicon, metal, underfill material, molding compound, and/or similar such material. One or more portions of the second underfill layer **1716** may be disposed between the cold plate **1702** and the package substrate **1706**. In some embodiments, one or more portions of the second underfill layer **1716** are adjacent to the cold plate **1702**, the device **1704**, the first underfill layer **1710**, and/or the interposer **1712**. In some embodiments, the second underfill layer **1716** comprises one or more edges. The one or more edges may comprise a sloped profile. For example, a first edge **1718** of the second underfill layer **1716** may have a sloped profile. In some embodiments, the sloped profile of one or more edges of the first underfill layer **1710** is the same or similar to the sloped profile of one or more edges of the second underfill layer **1716**.

(133) FIGS. **18A-18E** show a method of manufacturing a portion of a device package that may be used with the system panel, in accordance with some embodiments of the present disclosure. In some embodiments, the described methodology may be used to manufacture one or more portions of a device package described herein.

(134) FIG. **18A** shows a device **1802** attached to an interposer **1804**. The device **1802** may be attached to the interposer **1804** using any of the methodologies described herein. The device **1802** may have an active side

electrically connected to the interposer **1804** by use of conductive bumps, which are encapsulated by an underfill layer **1806** disposed between the device **1802** and the interposer **1804**. The underfill layer **1806** may comprise a cured polymer resin or epoxy, which provides mechanical support to the conductive bumps and protects against thermal fatigue. FIG. **18B** shows the device **1802** and the interposer **1804** attached to a package substrate **1808**. The device **1802** and the interposer **1804** may be attached to the package substrate **1808** using any of the methodologies described herein.

(135) FIG. **18C** shows a cold plate **1810** being attached to the device **1802**. The cold plate **1810** may be attached to the device **1802** using any of the methodologies described herein. For example, the cold plate **1810** may be directly bonded to the device **1802** without an intervening adhesive. In some embodiments, one or more standoffs are attached to the interposer **1804** and/or the package substrate **1808** prior to the cold plate **1810** being attached to the device **1802**. The one or more standoffs may be attached using one or more adhesives. The one or more standoffs may be disposed adjacent to the device **1802**. For example, a first standoff **1812** may be attached to the interposer **1804** adjacent to a first side of the device **1802** and a second standoff **1814** may be attached to the interposer **1804** adjacent to a second side of the device **1802**.

(136) In some embodiments, one or more standoffs are attached to the cold plate **1810** prior to the cold plate **1810** being attached to the device **1802**. The one or more standoffs may be attached using one or more adhesives or direct bonding. For example, the first standoff **1812** may be directly bonded to a first portion of the cold plate **1810** and the second standoff **1814** may be directly bonded to a second portion of the cold plate **1810**.

(137) FIG. **18D** shows a mold compound **1816** being deposited on the portion of the device package shown in FIG. **18C**. FIG. **18E** shows a manifold **1818** being attached to the cold plate **1810**. In some embodiments, the manifold **1818** is attached to the cold plate **1810** using one or more adhesives. In some embodiments, the manifold **1818** is attached to the cold plate **1810** using one or more compression members (e.g., O-rings or gaskets). FIG. **18E** also shows the a package cover **1820** being attached to the manifold **1818**. The package cover **1820** may be attached to the manifold **1818** using any of the methodologies described herein. In some embodiments, the package cover **1820** is attached to the manifold **1818** using one or more fasteners. For example, compression screws may be used to secure a portion of the package cover **1820** to a PCB.

(138) FIGS. **19A-19D** show a method of manufacturing a portion of a device package that may be used with the system panel, in accordance with some embodiments of the present disclosure. In some embodiments, the described methodology may be used to manufacture one or more portions of a device package described herein.

(139) FIG. **19A** shows a device **1902** being attached to a cold plate **1910**. In some embodiments, the device **1902** is bonded to the cold plate **1910** using direct bonding. For example, a first substrate comprising the cold plate **1910** may be directly bonded to a second substrate comprising the device **1902** without an intervening adhesive. In some embodiments, one or more methodologies described in relation to block **62** and block **64** (shown in FIG. **6**) may be used to generate the structure displayed in FIG. **19A**. For simplicity, the following description is focused on forming one integrated cooling assembly comprising one cold plate **1910** and one device **1902**. However, as mentioned above, in some embodiments, a first substrate may comprise a plurality of cold plates and a second substrate may comprise a plurality of devices, such that a plurality of integrated cooling assemblies may be formed from the first and second substrates.

(140) FIG. **19B** shows the device **1902** being attached to a package substrate **1908**. The device **1902** may be attached to the package substrate **1908** using any of the methodologies described herein. The device **1902** may have an active side electrically connected to the package substrate **1908** by use of conductive bumps, which are encapsulated by an underfill layer **1905** disposed between the device **1902** and the package substrate **1908**. The underfill layer **1905** may comprise a cured polymer resin or epoxy, which provides mechanical support to the conductive bumps and protects against thermal fatigue.

(141) In some embodiments, one or more standoffs are attached to an interposer **1906** and/or the package substrate **1908** prior to the device **1902** being attached to the package substrate **1908**. For example, the first standoff **1912** may be attached to a first portion of the interposer **1906** and the second standoff **1914** may be attached to a second portion of the interposer **1906**. The one or more standoffs may be attached using one or more adhesives.

(142) In some embodiments, one or more standoffs are attached to the cold plate **1910** prior to the device **1902** being attached to the package substrate **1908**. The one or more standoffs may be attached using one or more adhesives or direct bonding. For example, the first standoff **1912** may be directly bonded to a first

portion of the cold plate **1910** and the second standoff **1914** may be directly bonded to a second portion of the cold plate **1910**. The one or more standoffs may be disposed adjacent to the device **1902**. For example, the first standoff **1912** may be attached to the cold plate **1910** adjacent to a first side of the device **1902** and the second standoff **1914** may be attached to the cold plate **1910** adjacent to a second side of the device **1902**.

(143) FIG. **19C** shows a mold compound **1916** being deposited on the portion of the device package shown in FIG. **19B**. FIG. **19D** shows a manifold **1918** being attached to the cold plate **1910**. In some embodiments, the manifold **1918** is attached to the cold plate **1910** using one or more adhesives. In some embodiments, the manifold **1918** is attached to the cold plate **1910** using one or more compression members (e.g., O-rings or gaskets). FIG. **19D** also shows a package cover **1920** being attached to the manifold **1918**. The package cover **1920** may be attached to the manifold **1918** using any of the methodologies described herein. In some embodiments, the package cover **1920** is attached to the manifold **1918** using one or more fasteners. For example, compression screws may be used to secure a portion of the package cover **1920** to a PCB.

(144) The embodiments discussed above are intended to be illustrative and not limiting. One skilled in the art would appreciate that individual aspects of the cooling assemblies, device packages, and methods discussed herein may be omitted, modified, combined, and/or rearranged without departing from the scope of the disclosure.

Claims

1. A device package comprising: a substrate; an interposer attached to the substrate; a first semiconductor device, wherein the first semiconductor device is attached to the interposer and the first semiconductor device has a first footprint; a cold plate attached to the first semiconductor device by direct dielectric bonds, wherein the cold plate has a second footprint larger than the first footprint; and a first standoff attached to the interposer, wherein the first standoff is adjacent to the first semiconductor device and the first standoff is between the cold plate and the interposer.
2. The device package of claim 1, wherein: a first portion of the cold plate is located outside the first footprint of the first semiconductor device; and the first standoff is between the first portion of the cold plate and the interposer.
3. The device package of claim 1, wherein: the cold plate comprises a first inlet, a first cavity, and a first outlet; and the first cavity extends along a portion of the cold plate.
4. The device package of claim 3, wherein: the first inlet is coupled to a first coolant line; the first cavity extends along the first semiconductor device; and coolant is disposed in the first cavity and directly contacts the first semiconductor device.
5. The device package of claim 3, wherein the cold plate further comprises a second cavity.
6. The device package of claim 3, wherein: the cold plate further comprises a base plate between the first cavity and the first semiconductor device; the first inlet is coupled to a first coolant line; the first cavity extends along the first semiconductor device; and coolant is disposed in the first cavity.
7. The device package of claim 3, wherein: the first inlet is coupled to a first coolant line; the first cavity extends along the first semiconductor device; and coolant flows from the first inlet through the first cavity and directly contacts the first semiconductor device.
8. The device package of claim 7, wherein the first inlet and the first outlet are outside of the first footprint of the first semiconductor device.
9. The device package of claim 1, wherein the first semiconductor device is a device stack.
10. The device package of claim 1, further comprising a second semiconductor device, wherein the second semiconductor device is attached to the interposer, and the cold plate is attached to the second semiconductor device.
11. The device package of claim 10, wherein the cold plate is attached to the first semiconductor device and the second semiconductor device by direct dielectric bonds.
12. The device package of claim 10, wherein: the second semiconductor device has a third footprint; and the second footprint is larger than a combination of the first footprint and the third footprint.
13. The device package of claim 1, wherein the first standoff comprises silicon.
14. The device package of claim 1, wherein the first standoff comprises metal.
15. The device package of claim 1, further comprising a second standoff attached to the interposer, wherein:

the second standoff is adjacent to the first semiconductor device; and the second standoff is between the cold plate and the interposer.

16. The device package of claim 15, wherein: a second portion of the cold plate is located outside the first footprint of the first semiconductor device; and the second standoff is between the second portion of the cold plate and the interposer.

17. A device package comprising: a substrate; an interposer attached to the substrate; a first underfill layer disposed on the interposer; a first semiconductor device, wherein the first semiconductor device is attached to the first underfill layer and the first semiconductor device has a first footprint; a cold plate attached to the first semiconductor device by direct dielectric bonds, wherein the cold plate has a second footprint larger than the first footprint; and a second underfill layer disposed on the interposer, wherein: a first portion of the second underfill layer is adjacent to the first semiconductor device; and the first portion of the second underfill layer is between the cold plate and the interposer.

18. A device package comprising: a substrate; an interposer attached to the substrate; a first semiconductor device, wherein the first semiconductor device is attached to the interposer and the first semiconductor device has a first footprint; a cold plate attached to the first semiconductor device by direct hybrid bonds, wherein the cold plate has a second footprint larger than the first footprint; and a first standoff attached to the interposer, wherein the first standoff is adjacent to the first semiconductor device and the first standoff is between the cold plate and the interposer.

19. A device package comprising: a substrate; an interposer attached to the substrate; a first underfill layer disposed on the interposer; a first semiconductor device, wherein the first semiconductor device is attached to the first underfill layer and the first semiconductor device has a first footprint; a cold plate attached to the first semiconductor device by direct hybrid bonds, wherein the cold plate has a second footprint larger than the first footprint; and a second underfill layer disposed on the interposer, wherein: a first portion of the second underfill layer is adjacent to the first semiconductor device; and the first portion of the second underfill layer is between the cold plate and the interposer.
